Technical Data Sheet

LOCTITE® LIOFOL LA 3610 / LA 6229

January 2024

Product description
LOCTITE® LIOFOL LA 3610 / LA 6229 provides the following product characteristics:

Technology	Polyurethane
Product type	Laminating adhesive
Condition	Solvent-based
Components	Two components – requires mixing
Application	Film/film and film/foil laminates
Performance	High
Cure	Room temperature cure after mixing
Appearance base LOCTITE® LIOFOL LA 3610	Yellow, transparent to slightly opaque
Appearance hardener LA 6229	Colourless to yellowish, clear
Mix ratio, by weight (base:hardener)	14 : 1

Application areas:
LOCTITE® LIOFOL LA 3610 / LA 6229 is suitable for film/film and film/foil laminates. It is recommended for the manufacture of retortable laminates with outstanding heat resistance. Laminates manufactured with suitable materials are resistant to boiling, pasteurisation and sterilisation (30 min, 121°C) (outer layer). The combination of thermic and mechanic stress - particularly when oriented films are used - has to be checked in each single case.

Suitable films:
- Aluminium foil
- Polyethylene
- Polypropylene
- Polyester
- Polyamide
- rigid PVC

Not suitable films:
- EVA-Copolymerfilms

Product properties:
LOCTITE® LIOFOL LA 3610 / LA 6229 meets the requirements of effective production techniques by increasing the machine speed and the channel temperature. The result is a fast cure and low contents of retaining solvents in the laminates. The cured adhesive film is transparent, odorless, elastic and age-resistant.

Technical data
LOCTITE® LIOFOL LA 3610:

Solids content, %	58 - 62
Viscosity, Brookfield - LVT, 20°C, mPa·s	
Spindle 2, speed 30 rpm	300 - 700
Volumetric density, g/cm³	1.1
Component	NCO component
Solvent	Ethylacetate

Technical data
LA 6229:

Solids content, %	100
Viscosity, Brookfield - LVT, 20°C, mPa·s	
Spindle 3, speed 60 rpm	700 - 1 700
Volumetric density, g/cm³	1.17
Component	OH component
Solvent	None

Compliance with food packaging regulations:
A food contact statement with further details is available on request. Under specific conditions LOCTITE® LIOFOL LA 3610 / LA 6229 is suitable for manufacturing of food packaging materials under the applicable FDA, EC-provisions and/or BfR recommendations.
LOCTITE® LIOFOL LA 3610 / LA 6229 contains aromatic isocyanates. We therefore recommend migration testing of the fully cured material under conditions of use, particularly with high temperature applications, to demonstrate compliance with the limits for primary aromatic amines. The responsibility for compliance with the specific conditions is with the converter and not with the adhesive supplier.

Directions for use

Preliminary statement:
Prior to use it is necessary to read the **Safety Data Sheet** for information about precautionary measures and safety recommendations. Also, for chemical products exempt from compulsory labeling, the relevant precautions should always be observed. Please also refer to the local safety instructions and contact Henkel for analytical support.

Protective measures:
As the adhesive contains polyisocyanates any contact with skin should be avoided. If skin gets in contact with LOCTITE® LIOFOL LA 3610 / LA 6229 please remove the adhesive immediately. Safety regulations with regard to handling of isocyanates have to be followed as per the guidelines issued by the International Isocyanates Institute.

Dilution:
Suitable solvents are ethyl acetate, acetone and Methylethylketone provided that the water content does not exceed 300 ppm.
Do not use aromatic or alcoholic solvents.

Dilution table
In order to obtain a certain solids content, the following quantities of solvent have to be added.
The Ford/DIN cup (4 mm/20 °C) viscosities are based on an average viscosity of LOCTITE® LIOFOL LA 3610 / LA 6229.

LOCTITE® LIOFOL LA 3610	LA 6229	Ethyl acetate	Solid content	Viscosity (sec.)
25 kg	1.79 kg	6.79 kg	50%	25 - 29
25 kg	1.79 kg	10.52 kg	45%	18 - 22
25 kg	1.79 kg	15.18 kg	40%	14 - 18
25 kg	1.79 kg	21.17 kg	35%	12 - 16
25 kg	1.79 kg	29.17 kg	30%	11 - 15

Mixing
LOCTITE® LIOFOL LA 3610 has to be diluted under stirring with corresponding solvent to the desired application solids. Then the correct amount of hardener LA 6229 has to be added under agitation.

Working life:
The ready-for-use adhesive in closed containers has a working time of about 8 hours without substantial viscosity increase. Ready-for-use adhesive which has not been consumed in the machine for more than 6 hours has to be disposed of. To be on the safe side, we suggest to dispose of the remaining adhesive in the machine unless it can be added gradually to fresh adhesive mixtures for non-critical laminates (no heat resistance, no chemical resistance).

Potlife under tropical conditions:
Please note that due to tropical conditions, i.e. high ambient temperatures and/or high humidity the potlife for the mix adhesive is shorter. For adhesive systems intended for high performance applications, not more adhesive should be mixed that can be consumed within a 2 hours time period, to be used directly after mixing.

Application:
LOCTITE® LIOFOL LA 3610 / LA 6229 can be applied with all dry laminating units with smooth or gravure rollers.

Application weight:
The usual application weight is 3 g/m² dry. Sterilizable and deep draw laminates require a coating weight of up to 4 - 5 g/m² dry. For printed laminates the weight has to be adjusted to higher weights accordingly, it should be approx. 1 - 2 g/m² higher than for unprinted films.

Working conditions:
High laminating temperatures together with a high line pressure of the calander improve the bond strength and optical appearance of the laminates.

Drying:
Drying conditions depend on the efficiency of the dryer and the line speed.

Curing:
Manufacture of triplex laminates is possible 2 - 4 hours after duplex production. The laminates may be rewound and slit after 1 - 2 days. Complete curing i. e. total resistance to severe environments will be achieved after 7 - 14 days of storage at room temperature depending on film combination and intended use of the laminates.

Film properties:
The completely cured LOCTITE® LIOFOL LA 3610 / LA 6229 polyurethane film is transparent, odourless, elastic and age resistant.

Storage
When stored in a dry and cool place in the original unopened containers, shelf life is as follows. Store separately from food. Once opened containers of LOCTITE® LIOFOL LA 3610 have to be closed tightly again and the material should be consumed within a short period. Opened containers of hardener LA 6229 can also be consumed later.

Shelf life

LOCTITE® LIOFOL LA 3610, months	9
LOCTITE® LIOFOL LA 6229, months	9

Classification:
Please refer to the corresponding **Safety Data Sheet** for details on:
Hazardous Information
Transport Regulations
Safety Regulations

Additional information

Film requirements
Film additives - particularly slip agent content in LDPE, MDPE and CPP films - packed goods, printing inks, pre-treatment and coating of films are of significant importance to the intended use of the laminates and may affect the performance properties either straight away or later.
Therefore, practical laminating tests as well as checks on the laminates are necessary before going into full scale production.
Further detailed information can be obtained from our "Technical Guide Line for Laminating with Liofol Laminating Adhesives".

Product range
Apart from LOCTITE® LIOFOL LA 3610 / LA 6229 we can offer the following products for the manufacture of film/foil laminates:

- Solventbased, solventless and waterbased laminating adhesives for flexible packaging and special applications
- Adhesives for high gloss lamination
- Primer for extrusion and PVDC-coating
- Coldseal adhesives
- Heatseal coatings
- Special cleaning agents

To obtain product specifications please ask for the relevant technical datasheets.

Disclaimer
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